MILITARY SPECIFICATION

CEMENT, SILICATE AND ZINC PHOSPHATE, DENTAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers silico-phosphate dental cement suitable for use in cementing fused porcelain restorations and orthodontic appliances and as a temporary posterior restoration.

1.2 Classification. Silico-phosphate cement shall be furnished in the following shades, as specified:

Shade No. 51 Pale Yellow
Shade No. 55 Light yellow

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.
PPP-C-186	Container, Packaging and Packing For Drugs, Chemicals and Pharmaceuticals.

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Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Millitary

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN CHEMICAL SOCIETY (ACS) PUBLICATIONS

Reagent Chemicals, ACS Specifications

(Application for copies should be addressed to the American Chemical Society Publications, 1155 Sixteenth Street, N. W., Washington, D. C. 20036).

3. REQUIREMENTS

3.1 Materials. The materials shall be of the highest quality throughout, and suitable in all respects for the purpose intended.

3.1.1 Powder. The powder shall be free from extraneous material such as dirt or lint. The pigment shall be uniformly dispersed throughout the powder.

3.1.2 Liquid. The liquid shall be water - clear and free from suspended material, and there shall be no deposits or sediment on the inside of the container.

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3.2 Description. The silico-phosphate dental cement shall consist of a glass bottle containing a minimum of 1/2 ounce of powder, a polyethylene drop dispensing bottle containing a minimum of 10 milliliters of liquid, and a set of instructions. Shall be suitable for the cementation of porcelain jacket crowns and orthodontic bands, and as a temporary restoration in posterior teeth. Shall be furnished in two shades. Shade No. 51 shall be pale yellow, and Shade No. 55 shall be light yellow (see 1.2).

3.3 Unset cement. Cements when spatulated in accordance with manufacturer's instructions shall not form lumps or granules, or evolve gas.

3.4 Color. The shades of color of the silico-phosphate cement shall conform to the color reflectance values for L, a, and b listed in table I, for the applicable shade number when tested as specified in 4.4.9.

Table I. Color Reflectance

Shade number	Color	Color Reflectance (Tolerance of 2.5 △ E units) L	a	b
51	Pale yellow	74.80	-1.2	+13.7
55	Light yellow	72.77	-0.6	+21.5

3.5 Physical properties. The silico-phosphate cement shall conform to physical properties listed in table II and as herein specified.

Table II. Physical Properties

Properties when used as below:	Testing Consistency of Mix Disk Diameter (min)	Time of Setting at 37° C (99° F) minutes (min)(max)	Compressive strength after 24 hours (min)	Film thickness microns (max)	Opacity C 0.70 (min)(max)	Solubility and Disintegration in 24 hours (max)	Arsenic content (max)
Cementing media	25 ± 1	5 - 8	1400	25	0.35 to 0.90	1.5	0.0002
Temporary posterior filling	25 ± 1	3 - 6	1700	-	-	1.0	0.0002

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3.6 Stability. The cement shall comply with all the requirements of this specification for no less than two years when stored in a cool, dry place (59 to 86 degrees Fahrenheit).

3.7 Delivery schedule. No more than three months shall have elapsed from date of manufacture of the cement to date of delivery to the Government.

3.8 Instructions. Adequate and accurate instructions for mixing the silico-phosphate cement shall accompany each unit package. Instructions shall include the following minimum requirements:

a. The powder-liquid ratio and testing consistency, in terms of grams of powder (± 5 percent) to milliliters (ml.) of liquid.

b. Temperature of the mixing slab and ambient temperatures.

c. Procedure of mixing the silico-phosphate cement both for use in cementing procedures and for use in filling procedures.

d. Total working time (i.e. mixing and seating time).

3.9 Workmanship. Workmanship shall be first class throughout. Powders, liquids, and their containers shall be free from defects which may detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in table III (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. Unit of product for examination purposes shall be one package of silico-phosphate cement consisting of one bottle of powder and one bottle of liquid.

4.2.2 Sampling for tests. Sufficient amount of silico-phosphate cement shall be taken at random from each lot or batch, and tested in accordance with table III.

Table III. Tests

Characteristic	Reqmt.	Test procedure	Unit of product to be tested per lot or batch number	Number of determinations per unit of product	Total number of determinations 3/
Testing consistency	3.5	4.4.2	1	3 1/	3
Time of setting	3.5	4.4.3	1	3 2/	3
Compressive strength	3.5	4.4.4	1	5 1/	5
Opacity	3.5	4.4.7	1	1 2/	1
Solubility and Disintegration	3.5	4.4.6	1	2 1/	2
Arsenic content	3.5	4.4.8	1	1 2/	1
Color determination	3.4	4.4.9	1	2 1/	2
Film thickness	3.5	4.4.5	1	3 1/	3

1/ A lot shall be rejected if the average of the determinations per unit of product fails the test.
2/ A lot shall be rejected if any of the determinations per unit of product fails the test.
3/ Number of determinations represents quantity for testing as a cementing media. Same number of additional determinations are required for testing as a posterior filling material (where applicable).

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4.3 Examination. The silico-phosphate cement shall be examined to determine compliance with specification requirements.

4.4 Tests.

4.4.1 Preparation of test specimens. The preparation of all test specimens shall be conducted at a temperature of 24.0 to 25.0 C. and at a relative humidity between 45 and 55 percent. The powder-liquid ratio shall be determined by the consistency test. The following mixing technic shall be employed in the preparation of all test specimens. The glass slab shall be approximately 6 inches (in.) long and 3 in. wide. The spatula (made of material not corroded by the cement) and the slab shall be clean and free from hardened particles of cement. The mixing time shall be one minute. A mass of powder, approximately one half of the total amount, shall be drawn into the liquid and spatulated for 15 seconds. Another portion, approximately one fourth of the total amount, shall be incorporated in the following 15 seconds. The remaining one-quarter portion shall be incorporated in 15 seconds. Then the whole mass shall be spatulated for the remaining 15 seconds. Approximately one third of the top surface of the slab shall be used. No particles of powder or any unused liquid shall remain on the slab when the mixing is completed. A linear and not a rotary motion of the spatula shall be used, with the edge of the spatula sweeping approximately one half of the mixing area of the slab on each stroke.

4.4.2 Testing consistency. The type of apparatus for measuring consistency shall be essentially that shown in figure 1. This apparatus consists of two flat glass plates, a weight, a gauge plug and a glass tube (inside diameter approximately 10 mm.) which will deliver 0.50 ml. of mixed cement. The combined weight of the top plate and the weight shall be 2,500 Gm.

Trial amounts of powder shall be mixed with 0.40 ml. of liquid. Then 0.50 ml. of each mix shall be delivered from the glass tube onto a flat glass plate. Two minutes after the mixing is started, the top glass plate, weighing approximately 2? Gm. and the necessary additional weight required to total 2,500 Gm., shall be carefully lowered onto the soft cement. Trials shall be made until the average of the major and minor diameters of the elbowed mass of cement is 25 ± 1 mm. 10 minutes after starting the test. The average weight of powder used in three successful determinations rounded off to the nearest 0.1 gm. combined with 0.40 ml. of liquid, shall be by definition the testing consistency or amount of powder necessary to produce a mix of standard consistency.

4.4.3 Time of setting. A type of apparatus found suitable for determining the time of setting is shown in figure 2. A brass ring held at room temperature and approximately 4.8 mm. high 11.1 mm. outside diameter and 9.5 mm. internal diameter shall be placed on a flat glass plate and filled with cement of standard consistency.

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Two minutes after starting the mix, the specimen shall be transferred to an atmosphere of 100 percent relative humidity at 37° C. Two and a half minutes after beginning the mix, a standard Gillmore needle, weighing 454 Gm. and having a flat end 1.06 mm. in diameter shall be carefully lowered vertically on to the horizontal surface of the cement. This shall be repeated at 30-second intervals. The time of setting is the number of minutes elapsed from the starting of the mix to the time when the needle fails to make a perceptible circle on the surface of the specimen when allowed to rest thereon under its own weight. The setting time shall be reported to the nearest minute.

4.4.4 Compressive strength. The test specimens shall be cylinders 12 mm. high and 6 mm. in diameter. The ends of the specimens shall be flat and smooth and shall be parallel to each other and at right angles to the long axis of the cylinder.

An apparatus found convenient for forming these test cylinders is shown in figure 3. A cylindrical mold 12 mm. high and 6 mm. in diameter shall be placed on a flat glass plate and slightly overfilled with cement of standard consistency within two minutes after commencing the mixing. A second flat glass plate shall be pressed on top of the mold. The mold and plates shall be held firmly together with a small C clamp. All apparatus shall be at room temperature. The molds shall be made of hard rubber, glass, stainless steel or any other substance which will not react with the cement. The molds may be coated with a 3 percent solution of a microcrystalline wax (melting point 86-91° C.) in benzene. Three minutes after starting the mix, the mold and clamp shall be transferred to an atmosphere of 100 percent relative humidity at 37° C. (fig. 2). One hour later the ends of the cylinder shall be surfaced plane at right angles to the axis. The ends of the specimens may be ground flat by the use of a small amount of 200-mesh silicon carbide powder and water. The molds containing the specimens shall be drawn back and forth across a glass plate coated with the abrasive and water. They shall be rotated about one-fourth turn every few strokes. The test specimens shall be kept wet during the grinding.

After surfacing, the specimens may be removed from the mold by a screw jack (fig. 3) and shall be immersed in distilled water at 37° C. The time lapse between the starting of the mixing and the crushing shall be 24 hours. A small piece of blotting paper (approximately 0.5 mm.) wet with water shall be inserted between the ends of the specimen and the platens of the testing machine. The specimens shall be loaded at the rate of 225 $\pm$ 50 kg./min. and shall be kept wet during the test. The value for compressive strength shall be reported as the average of three or more from a lot of five specimens and shall be rounded off to the nearest 10 kg./cm^2. If the values for individual specimens fall more than 15 percent below the average of the five, they shall be discarded and the average of the remaining specimens shall be reported. In case more than two of the specimens are eliminated, the test shall be repeated.

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4.4.5 Film thickness. A portion of a mix of standard consistency shall be placed between two flat square or round glass plates of uniform thickness. The surface areas of the plates between which the cement is spread shall be approximately 2 square centimeters. Three minutes after the mix is started, a load of 15 kg. shall be applied vertically on the top plate. Ten minutes after the mix is started, the thickness of the two plates with the cement film between them shall be determined. The difference in the thickness of the plates with and without the cement film shall be considered as the film thickness. An average of three tests shall be reported to the nearest 5 microns.

4.4.6 Solubility and disintegration. One half ml. of cement of standard consistency shall be placed in a split stainless steel ring (inside diameter 20 mm. and 1.5 mm. thick) which is sitting on a thin polyethylene sheet backed by a flat plate. Another flat plate faced with a sheet of thin polyethylene shall be used to press the cement into the ring. A piece of fine platinum or corrosion-resistant wire shall be placed in the soft cement as the specimens are formed to provide a convenient means of holding the specimens. Three minutes after the mix is started, the plates and cement shall be placed for one hour in an atmosphere having a relative humidity of 100 percent at 37° C.

After one hour, the specimens shall be withdrawn from the oven and two specimens shall be placed in each tared weighing bottle and weighed (fig. 4). The combined weight of the two specimens and the weighing bottle, less the weight of weighing bottle and the platinum wire, shall be taken as the weight of the specimens of cement. The specimens shall immediately be submerged by pouring 50 ml. of distilled water into the weighing bottle, which shall be stored for 23 hours at 37° C. The specimens shall then be removed from the water. There shall be no evidence of crystal growth on the surface of the specimens. The water shall be evaporated from the weighing bottle at a temperature just below 100° C. The weighing bottle shall then be dried at 150° C. to constant weight. After cooling to room temperature in a desiccator, containing thoroughly dry anhydrous calcium sulfate ($CaSO_4$) or silica gel (freshly dried at 130° C.), the weighing bottle and contents shall be weighed with a precision of 0.2 mg. This cycle of heating the weighing bottle to 150° C., cooling over a desiccant and reweighing shall be repeated until the weight loss of each bottle is not more than 0.5 mg. in any 24-hour period. The difference between the final weight of the weighing bottle and its initial weight shall be the amount of disintegration. The gain in weight divided by the weight of the specimens times 100 gives the percentage of disintegration. The average of duplicate tests (two weighing bottles containing two specimens each) shall be reported to the nearest 0.1 percent.

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4.4.7 Opacity. The opacity shall be represented by the contrast ratio $C_{0.70}$. The contrast ratio, $C_{0.70}$, is the ratio between the daylight apparent reflectance of the cement specimen (1 mm. thick) when backed by a black backing and the daylight apparent reflectance of the specimen when backed by a white backing having a daylight apparent reflectance of 70 percent relative to magnesium oxide(MgO).

A sufficient amount of cement of standard consistency shall be pressed between two flat glass plates to form a disk of cement approximately 3 cm. in diameter and 1 $\pm$ 0.025 mm. thick. Three minutes after the mix is started, the plates and cement shall be placed for one hour in an atmosphere having a relative humidity of 100 percent at 37° C. The specimen shall then be removed from the plates and stored for 23 hours in distilled water at 37° C.

A comparison of the opacities of the cement specimen and two opal glass standards, with $C_{0.70}$ values of 0.35 and 0.90, respectively, shall be made by placing the specimen and standards against a variegated black and white background. A film of distilled water shall cover the cement specimens and the standards and also the space between them and the black and white backing during observations. If the opacity of the specimen is between or equal to either of the opacities of the standards, the cement shall comply with this requirement. Any instrument may be used to obtain the $C_{0.70}$ values providing the accuracy of the instrument is within $\pm$ 0.02 $C_{0.70}$.

4.4.8 Arsenic content. One gm. of hardened cement, obtained from a specimen 24 hours old that has been stored in a dry airtight container shall be powdered to pass a no. 200 sieve. The powdered sample shall be digested in 100 ml. of distilled water on a steam bath for one hour. The filtrate shall be used in the test for water-soluble arsenic.

The test for arsenic shall conform to that designated in Reagent Chemicals, American Chemical Society Specifications (2.2).

4.4.9 Color. The color reflectance determination shall be made on a Color-master Model V colorimeter with small aperture accessory in illuminant C condition. The values of L, a, and b, with a tolerance of 2.5 Δ E units, shall be as specified in table I for the specified shade number.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One package, containing one bottle of powder, one bottle of liquid of the shade specified, and one set of instructions, constitutes one unit of issue.

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5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	6 units	96 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Immediate container.

5.1.3.1.1 Liquid. Ten milliliters (minimum) of liquid shall be packaged in a polyethylene dispensing bottle approximately 2-1/4 inches high and 1 inch in diameter. The bottle shall be equipped with a friction plug dispensing tip and a plastic screw cap cover molded to cover dispenser. In addition, the cover shall be secured with a tamperproof aluminum seal.

5.1.3.1.2 Powder. One half ounce of powder (minimum) shall be packaged in a glass, wide mouth bottle approximately 2 inches high and 1-1/4 inches in diameter. The bottle shall be supplied with a plastic snap-on cap. The cap shall be secured with a tamperproof plastic band.

NOTE: Bottles. Closures and marking of bottles shall conform to the requirements specified in PPP-C-196.

5.1.3.2 Individual box. Each bottle of liquid and each bottle of powder shall be individually packaged in a box of appropriate size constructed in accordance with PPP-B-566 except that commercial colors are acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

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5.1.3.3 Unit package. One individual box containing a bottle of liquid and one individual box containing a bottle of powder along with instructions for use shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676 except that commercial colors are acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.3.4 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676 except that commercial colors are acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Special requirements of Table II of PPP-B-636 shall apply for fiberboard boxes.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 or PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II of PPP-B-636. Special requirements specified in Table I of PPP-B-636 shall apply for fiberboard boxes.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Immediate container. In addition to any commercial markings, each bottle of liquid and each bottle of powder shall be marked with name and address of the manufacturer, date of manufacture for powder or date of pack for liquid, lot (control) number, name of contents, and quantity of contents. When the manufacturer is not the contractor, the name and address of the contractor shall also be shown.

5.3.2 Individual box. Each box shall bear the same information as required for the immediate container.

5.3.3 Unit package and Intermediate package. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Lot (control) number, contract or purchase order number and name of contractor shall be shown. Date of manufacture shall be shown in lieu of the date packed. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE (59°-86° F)".

5.3.4 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number shall be shown. Date of manufacture shall be shown in lieu of the date packed. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE (59°-86° F)".

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Shade required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Limitations. This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are **commercially available, but is intended to cover the types which are normally procured to meet military requirements.**

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

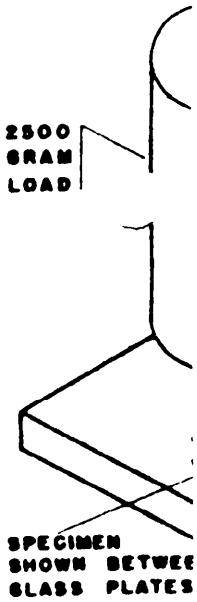
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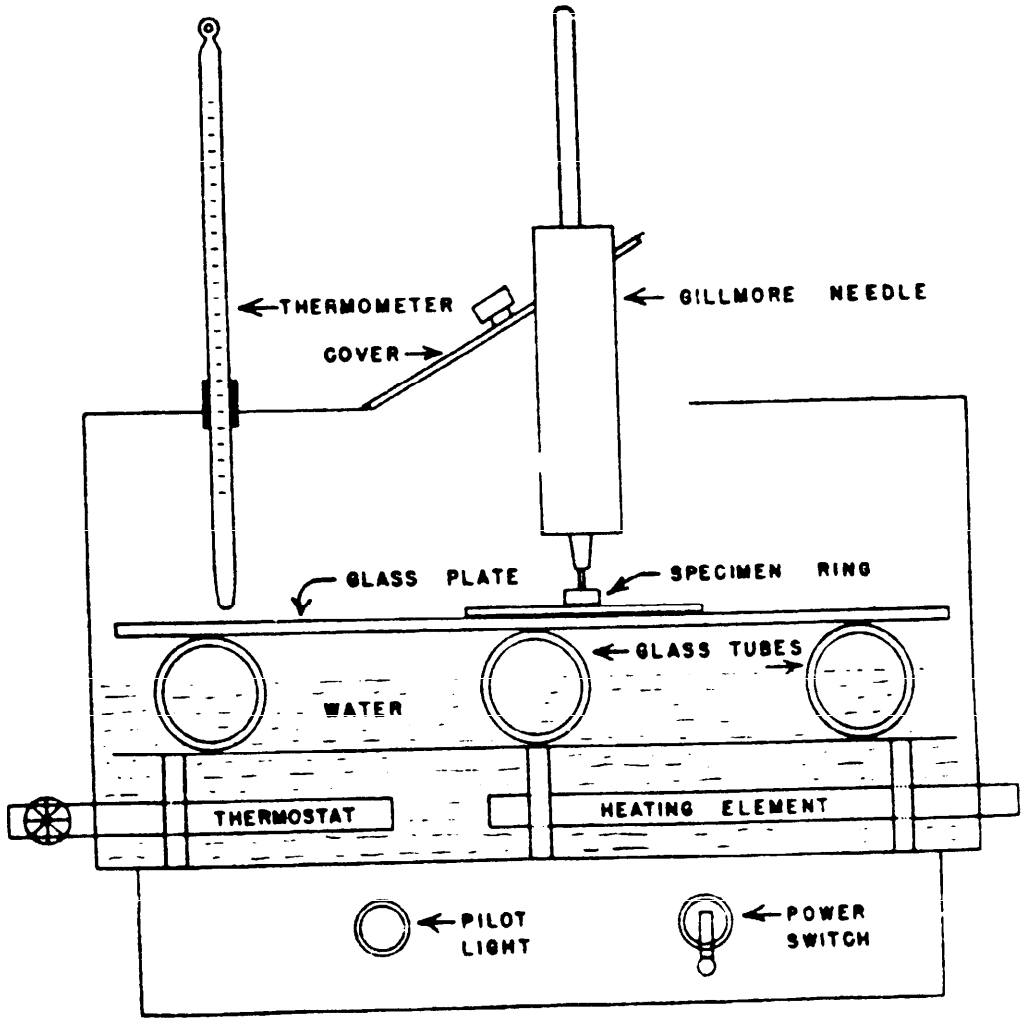
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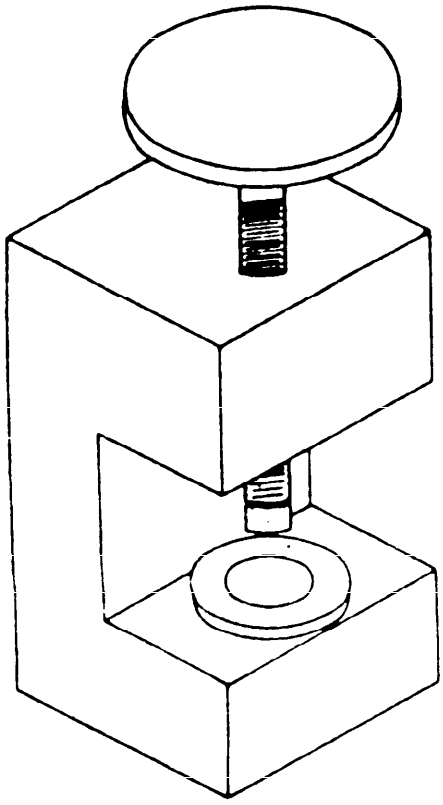


APPARATUS FOR DETERMINING SETTING TIME AT CONTROLLED TEMPERATURE AND HUMIDITY

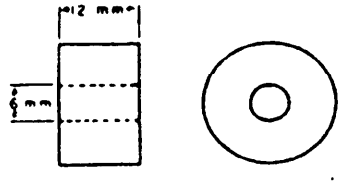
FIGURE 2

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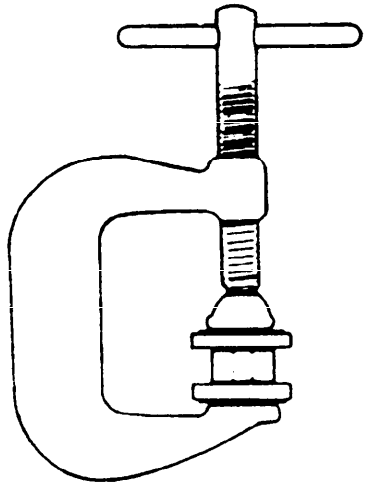
APPARATUS USED IN FORMING ULTIMATE COMPRESSIVE STRENGTH SPECIMENS



SCREW JACK FOR EJECTING SPECIMENS FROM MOLDS



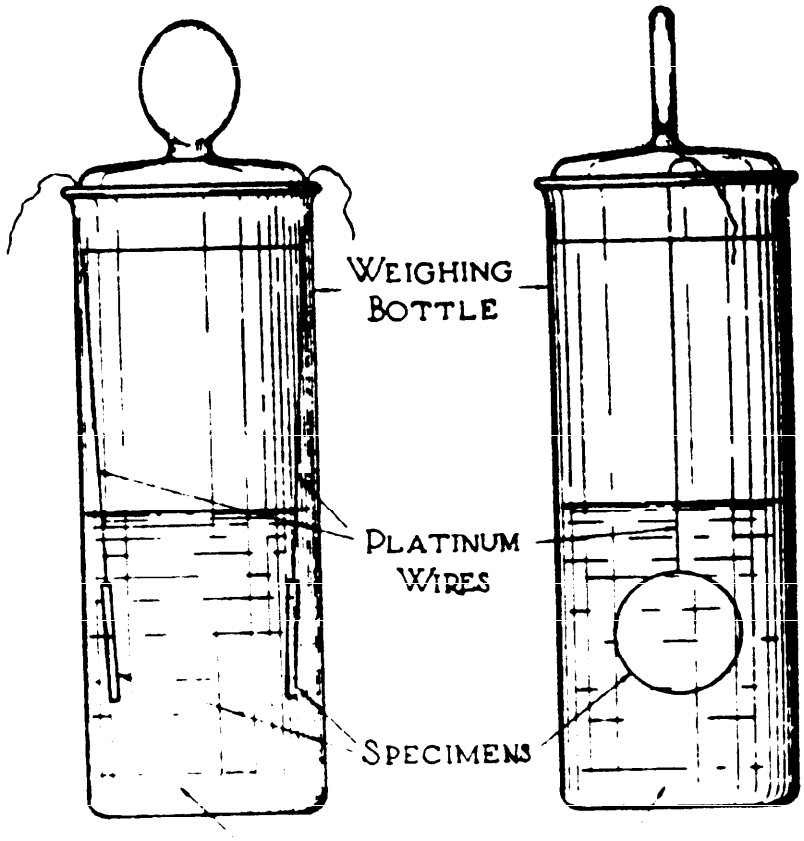
SPECIMEN MOLD



MOLD IN CLAMP

FIGURE 3

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WEIGHING BOTTLE CONTAINING SOLUBILITY SPECIMENS OF

SILICO - PHOSPHATE CEMENT

FIGURE 4

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements .

SPECIFICATION
MIL-C-36814

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

Fold

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2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

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HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATS
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PHILADELPHIA, PA. 19101

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DPSC FORM 2418
MAR 68

EDITION AUG 66, WILL BE USED UNTIL EXHAUSTED

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